

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 1 of 8

(1) Research Program/Project/Study Title:	Genetic Diversity and Population Dynamics of Philippine Native Pigs in Mindanao, Philippines
Proponent Name:	John King N. Layos - Instructor I
Academic Rank:	Instructor I
Co-Authors (with academic rank):	Marizel L. Santural - Instructor I Elsie L. Baranda - Instructor I Masahide Nishibori – Professor Takihiro Yonezawa – Professor Joy B. Banayo – Lab. Head
Campus/Satellite College/Address:	CAPSU Buriyas, Tapaz Campus, Hiroshima University, Tokyo University of Agriculture, UP Los Baños
Telephone/Cell Phone/Fax Number:	+639 274778024
E-mail Address:	Johnkinglayos5@gmail.com

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Mambusao						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☒ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Agriculture/Animal Science/Native Pigs		(8) Discipline/Field of Specialization Animal Science/Population Genetics							

	Document Type:	DOCUMENTED INFORMATION		Document Code	RES- RP-01
		ISO 9001:2015		Revision No.	00
	Document Title:	RESEARCH PROPOSAL		Effective Date	November 30, 2019
				Page	Page 2 of 8

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The Philippines is an archipelago of 7,641 islands situated in Island Southeast Asia (ISEA) tagged as a nexus of ancient human migrations within the western Pacific region (Larena et al., 2021). It is a mega-biodiverse country with almost half of the terrestrial vertebrates and vascular plants considered endemic (Posa et al., 2008). Together with Madagascar it shares the distinction as both a mega-diverse country and a global hotspot for biodiversity conservation (Mittermeier et al. 1999). In recent decades, a growing community of biogeographers, population geneticists, conservation biologists, and phylogeneticists has begun to focus on the archipelago and its diverse, endemic life forms as a model system to investigate a variety of conceptual questions related to evolutionary diversification (Heaney et al., 2005, Brown & Diesmos 2009, Oaks et al. 2013; Brown et al., 2013). Although the Philippines has always been characterized as a region of global priority for species conservation, the unforeseen threat of extinction of some animal genetic resources is well known (Heaney and Mittermeier 1997; Myers et al. 2000; Roberts et al. 2002; Posa et al. 2008).

The Philippines has one of the highest wild pig diversities in the world. It harbors four endemic wild pigs such as the Philippine warty pigs (*Sus philippensis*), Visayan warty pigs (*Sus cebifrons*), Palawan bearded pig (*Sus ahoenobarbus*), and Mindoro warty pig (*Sus oliveri*), as well as one native shared with Sundaic biogeographic region, the Bornean bearded pig (*Sus barbatus*) (Oliver, 1995). Unfortunately, these wild species are listed as Critically Endangered in the International Union for Conservation Nature (IUCN) Red List (Oliver, 2008). Although they do not receive much international attention, maintaining a viable population of these ecologically important species should be a high conservation priority.

It is interesting to note that the Philippines was never thought to be connected to the Asian continent, not even during the Quaternary sea-level subsidence (Voris, 2000). Thus, the faunal assemblages of the Philippines have become vital as it presents several palaeoecological, biogeographic, and archaeological questions and offers a unique evolutionary and ecological laboratory for understanding island biodiversity changes in Southeast Asia (Ochoa, 2019). For instance, the lack of archaeological evidence and molecular studies on *Sus scrofa* have poses a challenge in identifying its prehistoric arrival and domestication in the Philippines. The *S. scrofa* is a ubiquitous species that was not considered native to the Philippines and was likely introduced as a domestic animal within the last few thousand years (Ingicco et al., 2017). Studies have shown that *S. scrofa* has a very broad natural habitat and has been independently domesticated in different parts of the world (Larson et al., 2005; Giuffra et al., 2000; Groves, 1981) and has adapted to a variety of new environments in a relatively short evolutionary time frame (Frantz et al., 2016). A schematic profile of wild boar origin, dispersal and domestication across Eurasia has been well documented using mitochondrial DNA (mtDNA) from sequences of wild boar, domestic pigs, and ancient specimens worldwide (Larson et al., 2005; Bellwood et al., 2005; Larson et al., 2007; Wu et al., 2007). In addition, long-term gene flow between domestic pigs and wild boars during and after domestication has been well documented throughout Eurasia (Giuffra et al., 2000; Kijas and Andersson, 2001; Franz et al., 2015; Yang et al., 2017). Although archaeological and genealogical evidence suggests that domestication of pigs occurred independently at multiple sites in Northeast Asia (NEA) and on the Mainland Southeast Asia (MSEA) (Larson et al., 2010; Wu et al., 2007; Yang et al., 2011; Jin et al., 2012; Li et al., 2017), and despite the role of the hypothesized Austronesian human expansion in ISEA (Bellwood, 2005), the origin, dispersal, and domestication of pigs in the Philippines remain unclear. To date, the only potential domestic pigs identified in the archaeological record of the Philippines are from the Neolithic (4000-3000 cal. BP) and early Metal Age (3000-2000 cal. BP) site at Nagsabaran in Northern Luzon, which confirmed the clear distinction between the domesticated pig and the Philippine warty pigs (Pipper et al., 2009; Amano et al., 2013), which is associated with the Neolithic expansion into ISEA and Oceania by Austronesian-speaking populations (Larson et al., 2005). However, this has recently been questioned as there is no evidence of domestic pigs in Taiwan at a similarly early date, casting doubt on the possible Neolithic introduction of domestic pigs to the Philippines (Li et al., 2015).

The only leading theory postulates that the Philippine pig is a product of indiscriminate interbreeding between numerous domesticated endemic Philippine wild pigs and an introduced pig breed (Eusebio, 1969) that was able to survive and reproduce even with minimal human intervention. However, this hypothesis remains tentative due to the paucity of molecular studies to support this claim, as the evolution and dispersal of Philippine pigs have yet to be elucidated. Today, they are very common even in the remotest villages throughout the country. Since they are among the indigenous animals found in most rural agricultural areas, they are of great importance for supplementary income, high quality protein food, and socio-cultural and economic services, especially in cultural festivals and ceremonies. Therefore, genetic studies of the diversity of these ecologically and economically important animals should be a priority for conservation strategies, as they represent excellent genetic resources for local economies and could also serve as a genetic basis for studying human settlement and migration.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 3 of 8

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned).

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

This study aims to determine the genetic diversity, phylogeography, population dynamics, and extent of genetic introgression of Philippine native pigs in Mindanao Island using the mitochondrial DNA control region and to contribute important insight towards elucidating the history of dispersal and evolution of pigs worldwide.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of-the-art of current technology / information from which the project proposal will take-off).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Sample Collection and Ethical Approval

The experimental procedures will be conducted in accordance with institutional and national guidelines governing the care and use of animals in experiments. A total of 250 samples of Philippine native pigs (including wild pigs) in the Central, West, East and Northern Regions of Mindanao. Prior to the start of this study, communications will be sent to the concerned agencies (i.e., DA, DENR, BAI, LGU) for the proper coordination and assistance concerning the conduct of this study. Furthermore, since most of the sampling will be done in the remotest areas across Mindanao, the absence of pedigree record is assumed to be one of the limitations in this study. Therefore, to ascertain the unrelatedness of the samples, farmers will be interviewed and the Guidelines of Measurements of Domestic Animals Diversity Program set by the Food and Agriculture Organization (FAO, 2011) will be strictly implemented throughout the sampling procedure. Publicly available sequences of Philippine native pigs from Luzon and Visayas will also be included in the analysis for better resolution of the phylogenetic relationships and dynamics. Photographs will also be taken to document the morphological characteristics and differences within these pig populations. Finally, the animal owners will be personally consented to have their animals included in this study.

DNA Extraction, PCR Amplification and Sequencing

Genomic DNA will be extracted from whole blood and meat tissues of Philippine pigs using the phenol-chloroform method according to the recommended protocol of Green et al. (2012). For hair samples, the ISOHAIR kit (available at <http://www.nippongene.com>) will be used for DNA isolation.

The 5.0-kbp of mtDNA fragment will be first amplified with a Long and Accurate-PCR (LA -PCR) kit (KOD FX- Neo Polymerase, TOYOBO, Otsu, Japan) using the established primer set, forward: *Sus* mt. 5.0 FL -2: 5'-ATGAAAAATCATCGTTGTACTTCAACTACAAGAAC-3'; reverse: Mum R: 5'-TTCAGACCGACCGGAGCAATCCAGGTCGGTTTCTATCTA-3'. The reaction will start with an initial denaturation at 94°C for 2 min, followed by 30 cycles of denaturation at 98°C for 10 s, annealing at gradients at 57°C for 30 s, and primer extension at 68°C for 2 min and 30 s. The last step will be the final extension at 68°C for 8 min. For complete amplification of the mtDNA displacement (control region) region, approximately 1.3 kbp fragment will be amplified with another primer set, forward: *Sus* mtD F1: AACTCCACCATCAGCACCCAAAG; reverse: *Sus* mtD R1: CATTTTCAGTGCCTTGCTTTGATA. The PCR reaction will be performed in a total volume of 20 µL and the final concentrations of each component will be as follows: 4.5 µL ddH₂O, 10 µL 2× buffer, 4.0 µmol/L dNTPs, 0.3 µL of each primer (10 Pmol) (F & R primer), 0.6 µL KOD-FX Neo, and 0.5 µL genomic DNA. The reaction will start with an initial denaturation at 94°C for 2 min, followed by 30 cycles of denaturation at 98°C for 10 s, annealing at gradient 59°C for 30 s, and extension at 68°C for 30 s. The last step will be a final extension at 68°C for 5 min. Amplification will be performed using GeneAmp PCR System 9700 (Applied Biosystems, Foster City, CA, USA).

	Document Type:	DOCUMENTED INFORMATION		Document Code	RES- RP-01
		ISO 9001:2015		Revision No.	00
	Document Title:	RESEARCH PROPOSAL		Effective Date	November 30, 2019
				Page	Page 4 of 8

The PCR products from the segmental amplification will be purified with Exonuclease I (ExoI) and Shrimp Alkaline Phosphatase (SAP) to degrade the remaining PCR primers and dephosphorylate the remaining dNTPs, respectively. Then, the mtDNA D-loop fragments will be sequenced with 3130/3130xl Genetic Analyzers (Applied Biosystems, Foster City, CA, USA).

The DNA extraction will be conducted in the Laboratory of Animal Biotechnology, UPLB, Laguna, Philippines, while the DNA amplification and sequencing will be done in the Laboratory of Animal Genetics, Hiroshima University, Japan.

DNA Sequence Alignment

The complete sequences of the mtDNA control region will be assembled from the overlapping forward and reverse sequences using GeneStudio™ Professional, available at <http://www.genestudio.com>. Profile alignments of the sequenced data will be performed using the ClustalW algorithm (Thompson et al., 1994) as implemented in Molecular Evolutionary Genetics Analysis (MEGA) (Tamura et al., 2013) to generate refined and continuous sequences for each animal. The nucleotide sequences will be evaluated relative to the representative haplotypes of Asian domestic pigs under accession number AB041480 along with the complete worldwide sequences of domestic and wild boars. About 1044 bp of the complete mtDNA D-loop sequences will be aligned and edited until one highly variable tandem repeat motif (5'-CGTGCCTACA-3') remained, as the number of repeats is variable within individuals, indicating a high degree of heteroplasmy (Ghivizzani et al., 1993), and thus the repeat itself is not phylogenetically informative. The haplotype sequences will be then submitted to the GenBank National Center for Biotechnology Information (NCBI) databases.

Genetic Diversity and Phylogenetic Reconstruction

The diversity measures such as the number of polymorphic segregating sites, haplotype diversity, and nucleotide diversity will be estimated using DNA Sequence Polymorphism (DnaSP) 5.10 software (Librado and Rozas, 2009).

Two datasets will be assembled for the phylogenetic analyses. The first dataset will be the newly sequenced data from 184 animals used to study the genetic structure of the Philippine pig population. For the second dataset, the complete mtDNA control region sequences of global domestic and wild boars representing Asian and European pigs will be downloaded from GenBank to make further inferences about the relationships and demographic distribution of the Philippine pig populations in Mindanao. The Bayesian phylogenetic tree based on posterior probabilities will be constructed using the program MrBayes 3.2 (Ronquist et al., 2012). Two softwares such as the MEGA 7.0.26 (Tamura et al., 2013) and jModelTest (Posada, 2008) will be used to determine the best-fitted model of molecular evolution. Trees will be rooted with Warthog (*Phacochoerus africanus*; DQ409327). For each tree, two independent Markov chain Monte Carlo (MCMC) will be run for 5×10^7 chain length (second dataset), sampled every 5,000 generations. The first 10% of the sampled trees and estimated parameters of each dataset will be discarded as burn-in. To obtain sufficient convergence of log-likelihood values, a standard deviation of <0.05 will be considered. The phylogenetic tree using maximum likelihood (ML) will be also constructed and the consensus trees will be illustrated using FigTree 1.3.1. Using BioEdit ver.7.1 (Hall, 1999), the haplogroup-specific mutations will be examined further in all of the samples to justify the haplogroup assignment of each sequence.

Haplogroup and Geographic Classification

To obtain more detailed information about the genealogical relationship between haplotypes, a median-joining (MJ) networks (Bandelt, Forster, and Röhl 1999) will be constructed using PopArt 1.7 (Leigh and Bryan, 2015). This method calculates the net divergence of each taxon from all other taxa as the sum of the individual distances of variance within and between groups of all sequences. The nomenclatures described by Larson et al. (2005) with six clades (D1 to D6) including the newly proposed mitochondrial Southeast Asia (MTSEA) haplogroup (Tanaka et al., 2008), previously renamed D7 by Layos et al. (2022), will be used as a reference for clade notation. We will also perform network analysis on shorter sequences (510 bp) to accommodate the major representative haplotypes by partial mtDNA resolution used in the previous studies for a thorough haplotype representation of the different haplogroups and geographical locations.

Population Expansion Estimation and Demographic History Analysis

Deviations from selective neutrality will be estimated using Fu's (1997) F_s based on a coalescent simulation algorithm and Tajima's D statistical tests using Arlequin (Excoffier et al., 2005), and their significance will be tested over 1,000 coalescent simulations. The Fu's F_s test is very sensitive to demographic expansion, resulting in large negative F_s values, whereas the significant Tajima's D value could be a sign of population expansion and bottleneck (Tajima, 1989).

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 5 of 8

The past population dynamics will be examined with the Bayesian Skyline Plot (BSP) model (Drummond et al., 2005) with standard MCMC sampling procedures under HKY+G model of substitution (Hasegawa et al., 1985) with four gamma categories using BEAST v.2.6.3 (Bouckaert et al., 2014). The BSP represents changes in population size over time derived from mtDNA and assumed mutation rate. Analyses will be performed for the entire dataset and the predominant haplogroups D2 and D7 using a mutation rate of 1.36×10^{-8} (mutation rate per nucleotide site per year according to previous estimates for the D-loop of mammalian mtDNA; Pesole et al., 1999) using the strict clock. MCMC analysis will be performed for 5×10^7 generations. Independent runs (logs and trees) will be pooled using Log Combiner, discarding the first 10% burn-in and sampling parameter values every 5,000 generations. We will run the MCMC simulation twice independently for all datasets to ensure that the simulation converged at the same rate. Tracer v.1.7 will be used to confirm the correct convergence of the MCMC chain with an effective sample size (ESS) >200 in the log files and to visualize the dynamics of the effective population size over time.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1						Year 2					
	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
1. Field work												
2. Laboratory Analysis of samples (DNA Extraction, Sequencing, Molecular Analysis)												
3. Presentation of results and writing of manuscript												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

2 article publications
1 extension activity

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Department of Agriculture (DA), Bureau of Animal Industry (BAI), Biodiversity Conservation Organization, Faculty, Students

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 6 of 8

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
John King N. Layos	Study Leader	Agriculture/Population Genetics
Marizel L. Santural	Member	Animal Science
Elsie L. Baranda	Member	Animal Science
Masahide Nishibori	Collaborator	Agriculture/Animal Genetics
Takihiro Yonezawa	Collaborator	Agriculture/ Animal Genetics
Joy B. Banayo	Collaboration	Agriculture/ Animal Genetics

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours): 3 hrs/week

Program/Project Members (in hours): 2 hrs/week

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1	75,000
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		75,000
- Meal allowance during data gathering		10,000
- Professional fees (laboratory analysis and other related fees)		300,000
- Miscellaneous (for incidental expenses)		20,000
- Printing		5,000
C. CAPITAL OUTLAY		485,000
Total		

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

- Bellwood, P., and Dizon, E. (2005). The Batanes Archaeological Project and the 'Out of Taiwan' hypothesis for Austronesian dispersal. *Austronesian Stud*, 1(1): 1–32.
- Brown, R., and Diesmos, A. Philippines, Biology: In: Gillespie, R. and Clague, D. (Ed.), *Encyclopedia of Islands*. Berkeley: University of California Press (2009). p. 723-732.
- Brown, R.M., Siler, C.D., Oliveros, C.H., Esselstyn, J.A., Diesmos, A.C., Hosner, P.A., Linkem, C.W., Barley, A.J., Oaks, J.R., Sanguila, M.B., Welton, L.J., Blackburn, D.C., Moyle, R.J., Peterson, A.T., and Alcalá, A.C. (2013). Evolutionary processes of diversification in a model island archipelago. *Ann. Rev. Ecol. Evol. Syst.* 44(1): 411-435.
- Giuffra, E., Kijas, J.M.H., Amarger, V., Calborg, O., Jeon, J., -T., and Andersson, L. (2000). The origin of the domestic pig: Independent domestication and subsequent introgression. *Genetics* 154:1785-1991. [doi: 10.1093/genetics/154.4.1785](https://doi.org/10.1093/genetics/154.4.1785)
- Groves, C.P. (1981). *Ancestors for the Pigs, Taxonomy and Phylogeny of the Genus Sus*. Department of Prehistory Technical Bulletin 3, Australian National University Press, Canberra, Australia. p. 69.
- Frantz, L.A.F., Meijaard, E., Gongora, J., Haile, J., Groenen, M.A.M., and Larson, G. (2016). The evolution of Suidae. *Annu. Rev. Anim. Biosci.* 4(1): 61–85.
- Heaney, L.R., Walsh, J.S., and Peterson, A.T. (2005). The roles of geological history and colonization abilities in genetic differentiation between mammalian populations in the Philippine archipelago. *J. Biogeogr.* 32:229–47.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 7 of 8

- Heaney, L., and Mittermeier, R.A. (1997). The Philippines. In: Mittermeier RA, Mittermeier CG, Robles GP. Me (eds) Megadiversity: Earth's Biologically Wealthiest Nations. Monterrey (Mexico), CEMEX. p. 236-255.
- Ingicco, T., Piper, P.J., Amano, N., Paz, V.J., and Pawlik, A.F. (2017). Biometric differentiation of wild Philippine pigs from introduced *Sus scrofa* in modern and archaeological assemblages. *Int. J. Osteoarchaeol.* 27:768-784. [doi: 10.1002/oa.2592](https://doi.org/10.1002/oa.2592)
- Larena, M., Quinto, F.S., Sjödin, P., McKenna, J., Ebeo, C., Reyes, R., Casel, O., Huang, J.Y., Hagada, K.P., Guilay, D., Reyes, J., Allian, F.P., Mori, V., Azarcon, L.H., Manera, A., Terando, C., Jamero, L., Sireg, G., Manginsay-Tremedal, R., Labos, M.S., Vilar, R.H., Latiph, A., Saway, R.S., Marte, E., Magbanua, P., Morales, A., Java, I., Reveche, R., Barrios, B., Burton, E., Salon, J.C., Kels, M.J.T., Albano, A., Cruz-Angeles, R.B., Molanida, E., Granehall, L., Vicente, M., Edlund, H., Loo, J.H., Ho, T.Y.W., Reid, L., Malmström, H., Schlebusch, C., Lambeck, K., Endicott, P., and Jakobsson, M. (2021). Multiple migrations to the Philippines during the last 50,000 years. *Proc. Natl. Acad. Sci. USA.* 118(13): e2026132118. [doi: 10.1073/pnas.2026132118](https://doi.org/10.1073/pnas.2026132118)
- Larson, G., Dobney, K., Albarella, U., Fang, M., Smith, E.M., Robins, J., Lowden, S., Finlayson, H., Brand, T., Willerslev, E., Conwy, P.R., Andersson, L., and Cooper, A. (2005). Worldwide phylogeography of wild boar reveals multiple centers of pig domestication. *Science.* 307:1618-1621. [doi: 10.1126/science.1106927](https://doi.org/10.1126/science.1106927)
- Mittermeier, R.A., Myers, N., Robles, P.G., and Mittermeier, C.G. (1999). Hotspots: Earth's biologically richest and most endangered terrestrial ecosystems. Mexico City: CEMEX.
- Oaks, J.R., Sukumaran, J., Esselstyn, J.A., Linkem, C.W., Siler, C.D., Holder, M.T., and Brown, R.M. (2013). Evidence for climate-driven diversification? A caution for interpreting ABC inferences of simultaneous historical events. *Evolution* 67:991-1010. [doi: 10.1111/j.1558-5646.2012.01840.x](https://doi.org/10.1111/j.1558-5646.2012.01840.x)
- Oliver, W.L.R. (1995). The taxonomy, distribution and status of Philippine wild pigs. *IBEX J.M.E.* 3:26-32.
- Oliver, W.L.R., and Heaney, L. (2008). *Sus philippensis*. The IUCN Red List of Threatened Species. e.T21176A9245407. [doi: 10.2305/IUCN](https://doi.org/10.2305/IUCN)
- Ochoa, J. (2019). Island Biodiversity and Human Palaeoecology in the Philippines: A zooarchaeological study of Late Quaternary faunas (Doctoral thesis). [doi: 10.17863/CAM.46067](https://doi.org/10.17863/CAM.46067)
- Posa, M.R.C., Diesmos, A.C., Sodhi, N.C., and Brooks, T.M. (2008). Hope for threatened tropical biodiversity: Lessons from the Philippines. *Bioscience.* 58(3):231-240. [doi: 10.1641/B580309](https://doi.org/10.1641/B580309)
- Roberts, C.M. (2002). Marine biodiversity hotspots and conservation priorities for tropical reefs. *Science* 295:1280-1284. [doi: 10.1126/science.1067728](https://doi.org/10.1126/science.1067728)
- Voris, H.K. (2000). Maps of Pleistocene Sea levels in Southeast Asia: Shorelines, river systems and time durations. *J. Biogeogr.* 27:1153–1167. [doi: 10.1046/j.1365-2699.2000.00489.x](https://doi.org/10.1046/j.1365-2699.2000.00489.x)

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

JOHN KING N. LAYOS

PERSONAL INFORMATION

Academic Rank:	Instructor 1
Date of First Appointment:	March 11, 2019
Residential Address:	CAPSU Burias, Mambusao, Capiz

ACADEMIC BACKGROUND

Graduate

Doctorate Degree:	Doctor of Philosophy
Specialization:	Agriculture/Animal Genetics

	Document Type: DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
		Revision No.	00
	Document Title: RESEARCH PROPOSAL	Effective Date	November 30, 2019
		Page	Page 8 of 8

School:	Hiroshima University
Year Completed:	2022
Master's Degree:	Master of Science
Specialization:	Animal Science
School:	Visayas State University
Year Completed:	2018
College	
Course:	Bachelor of Science in Agriculture
Specialization:	Animal Science
School:	Capiz State University
Year Completed:	2016